A_f , A_r (axis)
W_f_r , W_f_l , W_r_r , W_r_l (wheels)
N_f_r , N_f_l , N_r_r , N_r_l (nuts)
C_f_r C_f_l , C_r_r , C_r_l (covers)
E_C (end. check)

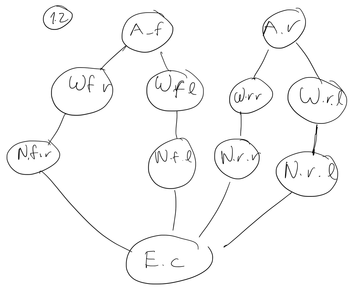
$A_f < (W_f_r \wedge W_f_l)$
$A_r < (W_r_r \wedge W_r_l)$

$W_f_r < N_f_r$ $W_f_l < N_f_l$
$W_r_r < N_r_r$ $W_f_l < N_f_l$

$N_f_r < C_f_r$ $N_f_l < C_f_l$
$N_r_r < C_r_r$ $N_f_l < C_f_l$

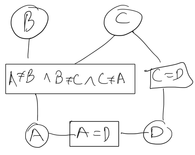
$(C_f_r \wedge C_f_l \wedge C_r_r \wedge C_r_l) < E_C$

12)



R1: Yes
R2: 2

3)



R1:
R2: Circular are Variables Rectangles are Constraints
R3: They only have the edges that the constraints need
R4: No $A=D \wedge D=C \wedge C \neq A \Rightarrow$ Impossible

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4.4.1

A_f , A_r (axis) — 10 min
W_f_r , W_f_l , W_r_r , W_r_l (wheels) — 1 min
N_f_r , N_f_l , N_r_r , N_r_l (nuts) — 2 min
C_f_r C_f_l , C_r_r , C_r_l (covers) — 1 min
E_C (end. check) — 3 min

$A_f < (W_f_r \wedge W_f_l)$
$A_r < (W_r_r \wedge W_r_l)$

$W_f_r < N_f_r$ $W_f_l < N_f_l$
$W_r_r < N_r_r$ $W_f_l < N_f_l$

$N_f_r < C_f_r$ $N_f_l < C_f_l$
$N_r_r < C_r_r$ $N_f_l < C_f_l$

$(C_f_r \wedge C_f_l \wedge C_r_r \wedge C_r_l) < E_C$

$A_f + 10 \leq (W_f_r \wedge W_f_l)$
$A_r + 10 \leq (W_r_r \wedge W_r_l)$

$W_f_r + 1 \leq N_f_r$ $W_f_l + 1 \leq N_f_l$
$W_r_r + 1 \leq N_r_r$ $W_r_l + 1 \leq N_r_l$
$N_f_r + 2 \leq C_f_r$ $N_f_l + 2 \leq C_f_l$
$N_r_r + 2 \leq C_r_r$ $N_r_l + 2 \leq C_r_l$
$C_f_r + 1 \leq E_C$ $C_f_l + 1 \leq E_C$
$C_r_r + 1 \leq E_C$ $C_r_l + 1 \leq E_C$

4.4.2

The minimum time will be the sum of all the tasks time so
10+10+1+1+1+2+2+2+2 + 1+1+1+1+3 = 39 minutes
CPS algorithms not always find the fastest solution, they algorith trys to find a viable solution that works for the constraints they were given.

4.4.3 10+1+2+1+3 = 17 min

The maximum nº of employees is 4 and they will complete it in 17 minutes

4.4.4 10+10+1+2+1+3 = 27 min

A_f , A_r (axis) — 10 min
W_f_r , W_f_l , W_r_r , W_r_l (wheels) — 1 min
N_f_r , N_f_l , N_r_r , N_r_l (nuts) — 2 min
C_f_r C_f_l , C_r_r , C_r_l (covers) — 1 min
E_C (end. check) — 3 min

$A_f < (W_f_r \wedge W_f_l)$
$A_r < (W_r_r \wedge W_r_l)$

$W_f_r < N_f_r$ $W_f_l < N_f_l$
$W_r_r < N_r_r$ $W_f_l < N_f_l$

$N_f_r < C_f_r$ $N_f_l < C_f_l$
$N_r_r < C_r_r$ $N_f_l < C_f_l$

$(C_f_r \wedge C_f_l \wedge C_r_r \wedge C_r_l) < E_C$

$A_f + 10 \leq (W_f_r \wedge W_f_l)$
$A_r + 10 \leq (W_r_r \wedge W_r_l)$

$W_f_r + 1 \leq N_f_r$ $W_f_l + 1 \leq N_f_l$
$W_r_r + 1 \leq N_r_r$ $W_r_l + 1 \leq N_r_l$
$N_f_r + 2 \leq C_f_r$ $N_f_l + 2 \leq C_f_l$
$N_r_r + 2 \leq C_r_r$ $N_r_l + 2 \leq C_r_l$
$C_f_r + 1 \leq E_C$ $C_f_l + 1 \leq E_C$
$C_r_r + 1 \leq E_C$ $C_r_l + 1 \leq E_C$

W1, W2, W3, W4 (Workers)

Worker_A.f = W1
Worker_A.r = W1
Worker_X ∈ {W1, W2, W3, W4} for other tasks

10 + 10 + 1 + 2 + 1 + 3 = 27 min (final inspection)

Worker_A.f — Worker_A.r — All (wheels) — All (nuts) — ALL (covers)